(5) Capstone Project: Full Alert-to-Response Cycle:

This capstone project simulates a complete SOC alert-to-response workflow. The objective is to understand how a security incident is handled from attack simulation to detection, triage, response, and reporting. Using tools such as Metasploit, Wazuh, CrowdSec, and Google Docs, the project demonstrates practical incident handling skills. The attack is launched against a vulnerable system, detected through monitoring, investigated using alerts, contained by blocking the attacker, and finally documented in both technical and non-technical reporting formats.

Lab set up:
Kali linux is a attacker machine
Metasploitable2 is victim
Ubuntu with wazuh and crowdsec is for monitor

Connect these 3 in same network NAT Network and check they can communicate with each other or not using ping command
From kali ,
ping 10.0.2.6
ping 10.0.2.4
From ubuntu,
ping 10.0.2.15
ping 10.0.2.6
From metasploitable2,
ping 10.0.2.15
ping 10.0.2.4

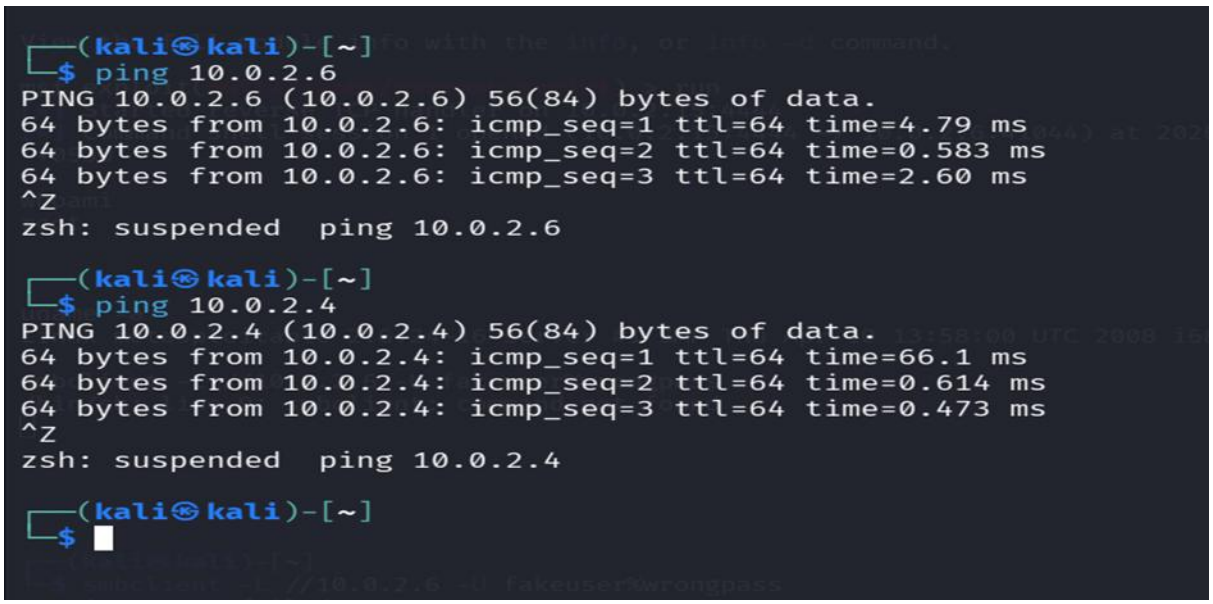

Figure 1: checking connection from kali

```
msfadmin@metasploitable:~$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=2.11 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=3.25 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=2.21 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=1.55 ms
64 bytes from 10.0.2.15: icmp_seq=5 ttl=64 time=3.23 ms

[33]+  Stopped                 ping 10.0.2.15
msfadmin@metasploitable:~$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=1.53 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=1.68 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=2.35 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=1.91 ms
64 bytes from 10.0.2.4: icmp_seq=5 ttl=64 time=1.74 ms
64 bytes from 10.0.2.4: icmp_seq=6 ttl=64 time=1.61 ms
64 bytes from 10.0.2.4: icmp_seq=7 ttl=64 time=2.33 ms
64 bytes from 10.0.2.4: icmp_seq=8 ttl=64 time=2.09 ms
64 bytes from 10.0.2.4: icmp_seq=9 ttl=64 time=1.76 ms
64 bytes from 10.0.2.4: icmp_seq=10 ttl=64 time=1.75 ms
z64 bytes from 10.0.2.4: icmp_seq=11 ttl=64 time=1.81 ms

[34]+  Stopped                 ping 10.0.2.4
msfadmin@metasploitable:~$ _
```

Figure 2: checking connection from metasploitable

```
vboxuser@UBUNTULINUX:~$ ping 10.0.2.6
PING 10.0.2.6 (10.0.2.6) 56(84) bytes of data.
64 bytes from 10.0.2.6: icmp_seq=1 ttl=64 time=14.1 ms
64 bytes from 10.0.2.6: icmp_seq=2 ttl=64 time=2.11 ms
64 bytes from 10.0.2.6: icmp_seq=3 ttl=64 time=2.00 ms
^Z
[1]+  Stopped                 ping 10.0.2.6
vboxuser@UBUNTULINUX:~$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=1.53 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=1.80 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=1.95 ms
^Z
[2]+  Stopped                 ping 10.0.2.15
vboxuser@UBUNTULINUX:~$
```

Figure 3: checking connection from ubuntu

In metasploitable2 configure syslog.conf to forward logs to ubuntu

sudo nano /etc/syslog.conf

add this at the end

. @ubuntu_ip

. @10.0.2.4

Then restart

sudo /etc/init.d/sysklogd restart

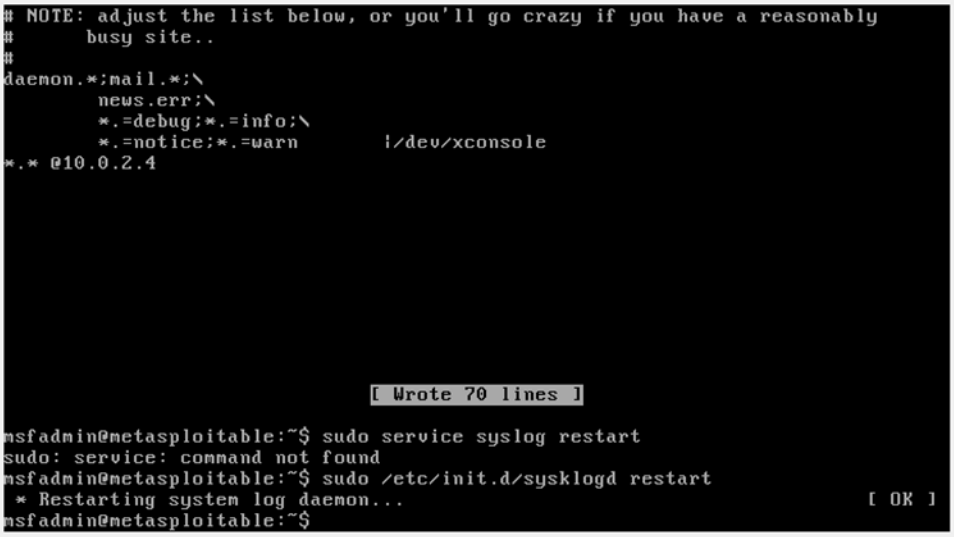

Figure 4: configuring metasploitable for forward logs to ubuntu

Remove comments from below lines in rsyslog.conf

module(load="imudp")

input(type="imudp" port ="514")

```
GNU nano 7.2                    /etc/rsyslog.conf *
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#
# Default logging rules can be found in /etc/rsyslog.d/50-default.conf


#################
#### MODULES ####
#################

module(load="imuxsock") # provides support for local system logging
#module(load="immark")  # provides --MARK-- message capability

# provides UDP syslog reception
module(load="imudp")
input(type="imudp" port="514")

# provides TCP syslog reception
#module(load="imtcp")
#input(type="imtcp" port="514")

# provides kernel logging support and enable non-kernel klog messages
module(load="imklog" permitnonkernelfacility="on")

###########################
#### GLOBAL DIRECTIVES ####
###########################

# Filter duplicated messages
$RepeatedMsgReduction on

#
```

Figure 5: configuration of rsyslog.conf file

On ubuntu confirm ubuntu receiving logs

sudo tail -f /var/log/syslog

test logs from metasploitable2 ,

logger “TEST LOG FROM METASPLOITABLE”

```
#
daemon.*;mail.*;\
        news.err;\
        *.=debug;*.=info;\
        *.=notice;*.=warn       |/dev/xconsole
*.* @10.0.2.4
msfadmin@metasploitable:~$ sudo /etc/init.d/sysklogd restart
 * Restarting system log daemon...                                    [ OK ]
msfadmin@metasploitable:~$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=1.11 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=1.20 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=1.34 ms

[5]+  Stopped                 ping 10.0.2.4
msfadmin@metasploitable:~$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=4.20 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=1.29 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=1.33 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=1.66 ms

[6]+  Stopped                 ping 10.0.2.15
msfadmin@metasploitable:~$ logger "TEST LOG FROM METASPLOITABLE"
msfadmin@metasploitable:~$
```

Figure 6: from metasploitable2 generating test log

From ubuntu test log is received from metasploitable2

```
2026-03-21T05:10:05.963882+00:00 UBUNTULINUX systemd[1]: Finished sysstat-collect.servi
ce - system activity accounting tool.
2026-03-21T05:10:33.496525+00:00 10.0.2.6 /USR/SBIN/CRON[4831]: (root) CMD (  [ -x /usr
/lib/php5/maxlifetime ] && [ -d /var/lib/php5 ] && find /var/lib/php5/ -type f -cmin +$
(/usr/lib/php5/maxlifetime) -print0 | xargs -r -0 rm)
2026-03-21T05:11:19.384292+00:00 10.0.2.6 dhclient: DHCPREQUEST of <null address> on et
h0 to 10.0.2.3 port 67
2026-03-21T05:11:19.437500+00:00 10.0.2.6 dhclient: DHCPACK of 10.0.2.6 from 10.0.2.3
2026-03-21T05:11:19.450281+00:00 10.0.2.6 dhclient: can't create /var/lib/dhcp3/dhclien
t.eth0.leases: Permission denied
2026-03-21T05:11:19.450281+00:00 10.0.2.6 dhclient: bound to 10.0.2.6 -- renewal in 257
 seconds.
2026-03-21T05:11:31.422923+00:00 10.0.2.6 msfadmin: TEST LOG FROM METASPLOITABLE
```

Figure 7: test log is received from metasploitable2

Create wazuh rules :
Create alert rules for wazuh in ubuntu, open custom rules file

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```

in this file add below rules

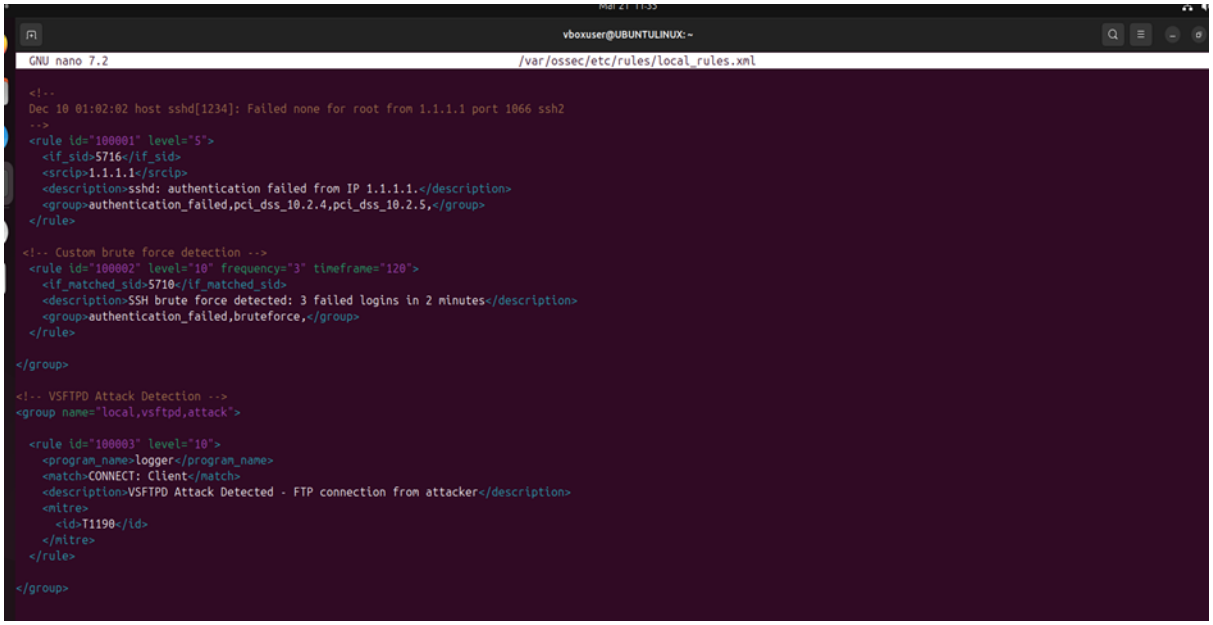

Figure 8: adding custom rules for wazuh

Ossec.conf file configuration:

In ossec.conf file

```
sudo nano /var/ossec/etc/ossec.conf
```

add ,

```
<localfile>
        <log_format>syslog</log_format>
        <location>/var/log/*.log</location>
</localfile>
```

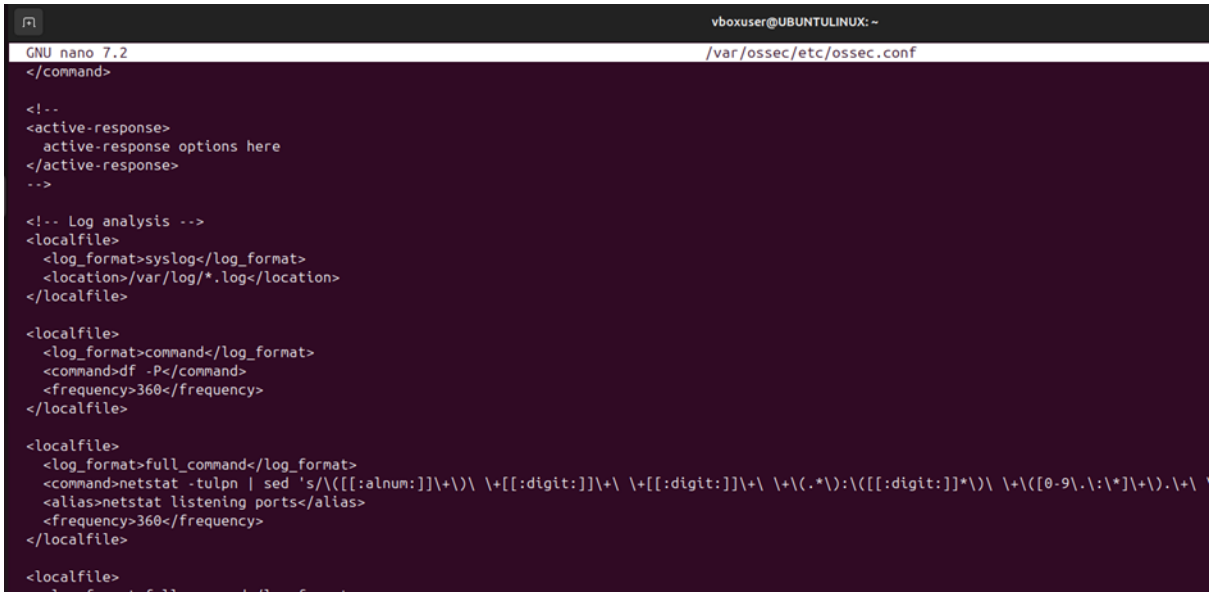

Figure 9: configuration of ossec.conf file

After that restart wazuh

sudo systemctl restart wazuh-manager

Attack Simulation:

In ubuntu,

sudo tail -f /var/log/syslog

In metasploitable

sudo tail -f /var/log/vsftpd.log | logger

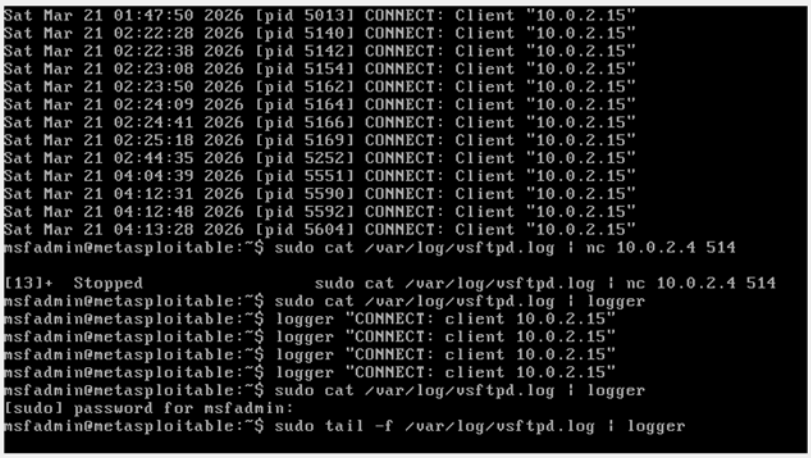

Figure 10: in metasploitable running sudo tail -f /var/log/vsftpd.log | logger

In kali linux simulate attack using Metasploit , in terminal start Metasploit by
Msfconsole

then run ,

use exploit/unix/ftp/vsftpd_234_backdoor
set RHOST 10.0.2.6
run

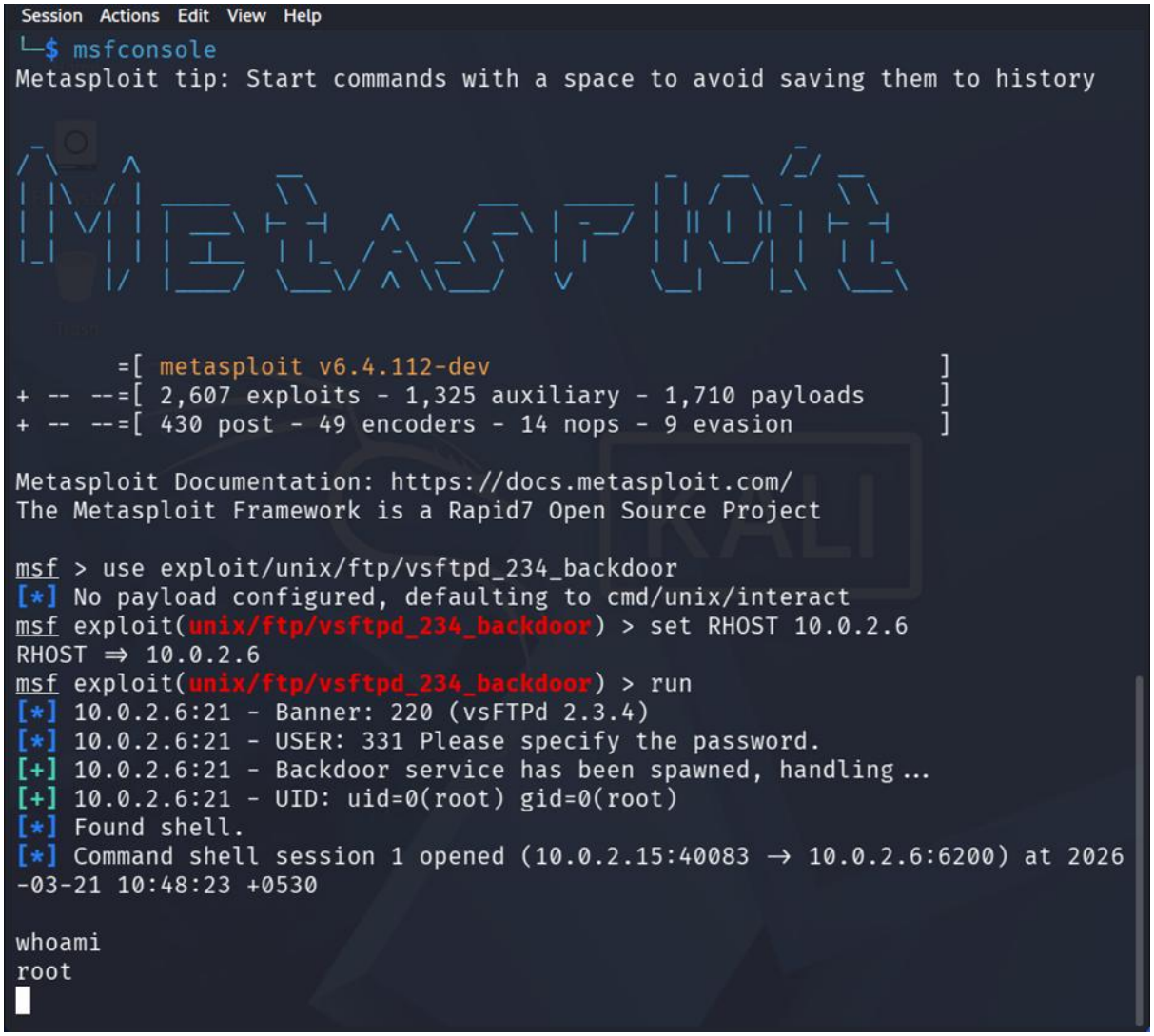

Figure 11: In kali linux simulated attack using Metasploit

Using Metasploit exploited the vsftpd vulnerability in metasploitable2. Now in ubuntu check logs from metasploitable2 arrived or not

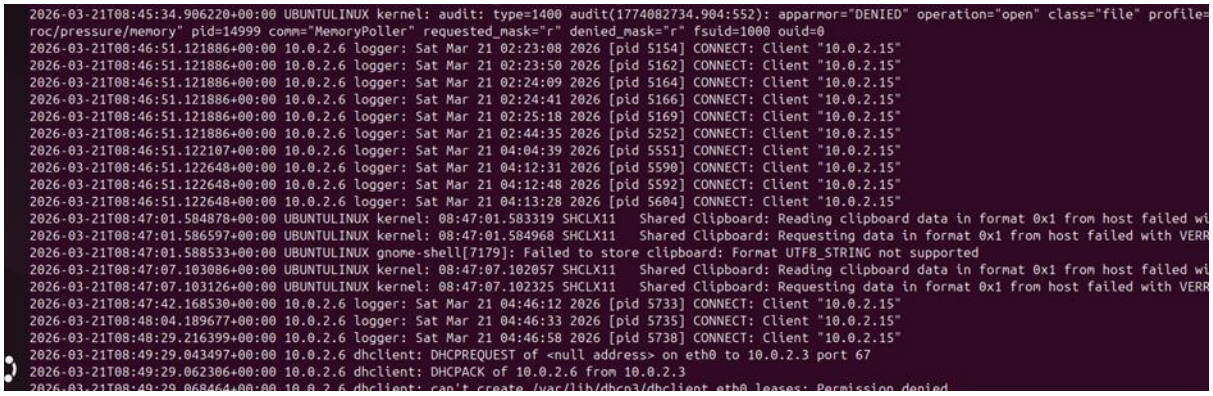

Figure 12: logs from metasploitable2

Detection and Triage:

To detect and analyze alert in wazuh goto browser

https://10.0.2.4:443

Goto wazuh dashboard → discover → in search bar type rule id and also after that goto threat hunting → events →in search bar type rule ide there .

rule.id=100003

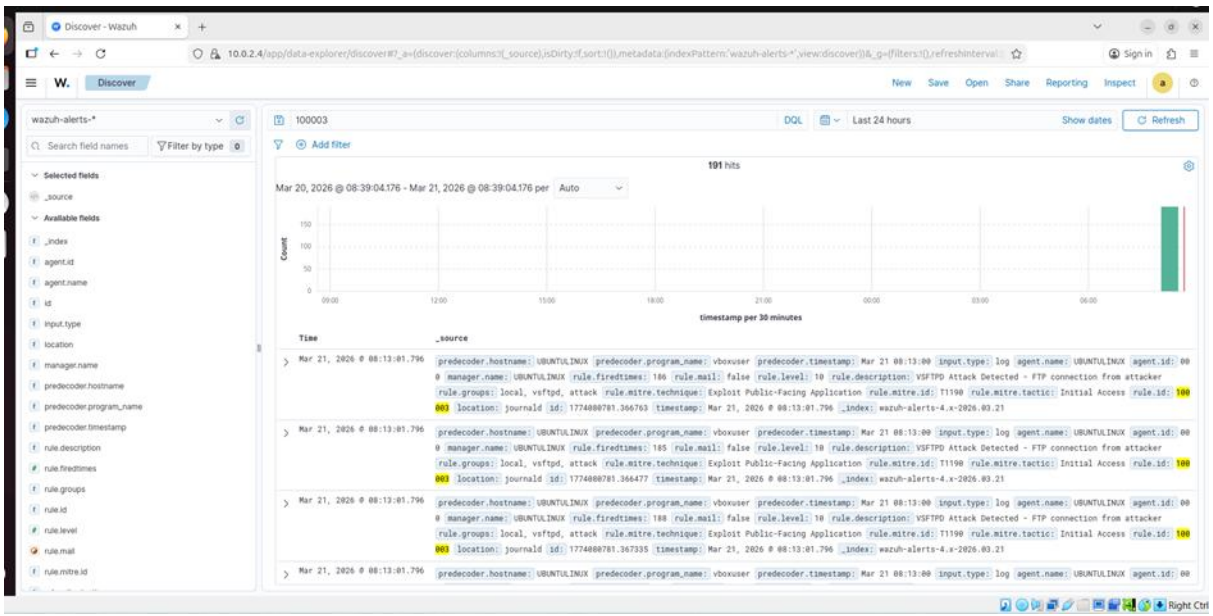

Figure 13: alerts are in wazuh discover

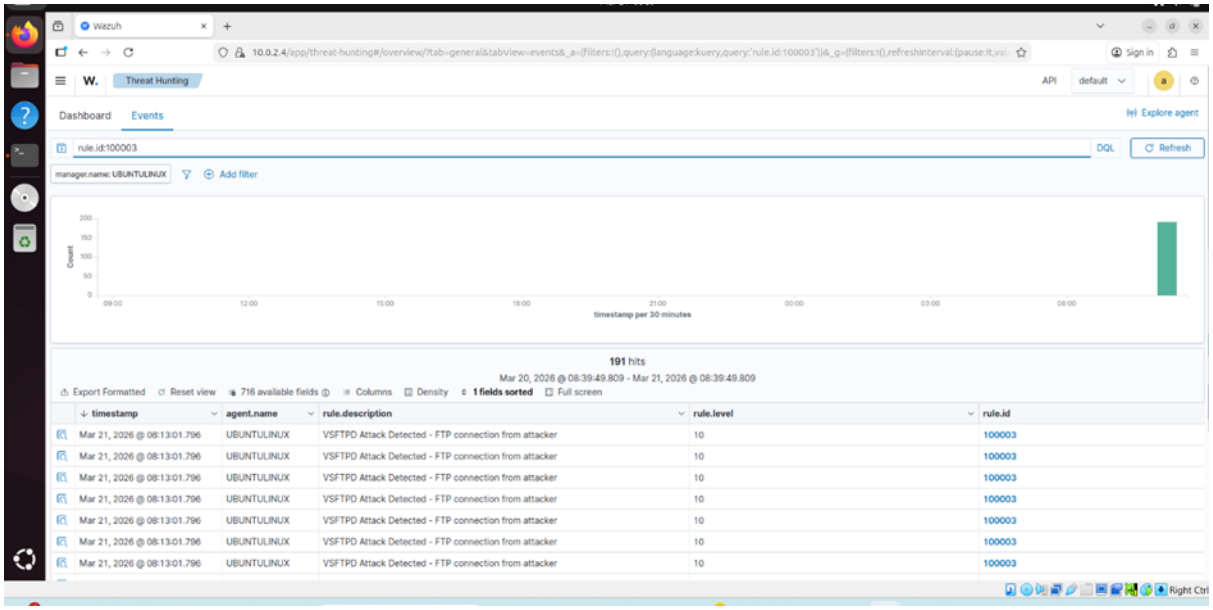

Figure 14: vsftpd alerts detected in wazuh

On march 21,2026@08:13:01.796 detected a vsftpd attack on metasploitable from Source ip 10.0.2.15. MITRE technique is T1190.

Document:

Time stamp	Source ip	Alert description	MITRE Technique
March 21,2026 08:13:01.796	10.0.2.15	VSFTPD Attack Detected - FTP connection from attacker	T1190

On March 21, 2026, at 08:13, a VSFTPD attack was detected from source IP 10.0.2.15. The alert indicated an unauthorized FTP connection attempt, mapped to MITRE technique T1190 Exploit Public-Facing Application. This event highlights a potential exploitation attempt against the server, requiring immediate investigation and response to prevent compromise.

Response:

Isolating the VM and blocking the attacker's IP using CrowdSec and verifying with a ping test.

Install Crowdsec on ubuntu
Add CrowdSec repository
sudo apt install -y curl gnupg lsb-release
curl -s https://packagecloud.io/install/repositories/crowdsec/crowdsec/script.deb.sh | sudo bash

Install crowdsec
sudo apt install crowdsec

start crowdsec
sudo systemctl start crowdsec

```
vboxuser@UBUNTULINUX:~$ sudo apt install crowdsec
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  crowdsec
0 upgraded, 1 newly installed, 0 to remove and 19 not upgraded.
Need to get 36.1 MB of archives.
After this operation, 120 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 crowdsec amd64 1
.4.6-6ubuntu0.24.04.2 [36.1 MB]
Fetched 36.1 MB in 16s (2,292 kB/s)
Selecting previously unselected package crowdsec.
(Reading database ... 351163 files and directories currently installed.)
Preparing to unpack .../crowdsec_1.4.6-6ubuntu0.24.04.2_amd64.deb ...
Unpacking crowdsec (1.4.6-6ubuntu0.24.04.2) ...
Setting up crowdsec (1.4.6-6ubuntu0.24.04.2) ...
I: Registering to LAPI (/etc/crowdsec/local_api_credentials.yaml)
I: Registering to CAPI (/etc/crowdsec/online_api_credentials.yaml)
I: Setting up offline hub (see README.Debian)
I: Enabling upstream-recommended items, first installation (via symlinks from /etc/crow
dsec)
I: Enabling WAL for SQLite [fstype=ext4] (see README.Debian)
Created symlink /etc/systemd/system/multi-user.target.wants/crowdsec.service → /usr/lib
/systemd/system/crowdsec.service.
W: Restarting manually to force a CAPI pull (upstream #2120)
vboxuser@UBUNTULINUX:~$ sudo systemctl start crowdsec
```

Figure 15: installed crowdsec on ubuntu successfully

Install the firewall bouncer
sudo apt update
sudo apt install crowdsec-firewall-bouncer-iptables

start and enable,

sudo systemctl enable crowdsec-firewall-bouncer

sudo systemctl start crowdsec-firewall-bouncer

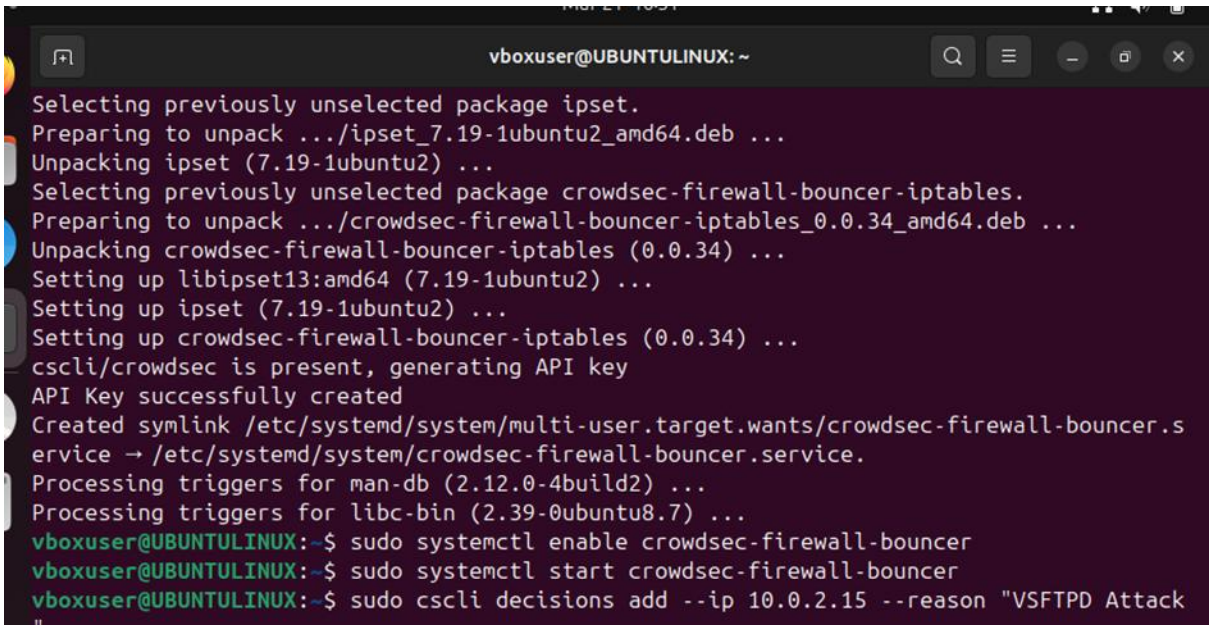

Figure 16: installed crowdsec

Now block the attacker IP using crowdsec,

add attacker ip

sudo cscli decisions add --ip 10.0.2.15 --reason "VSFTPD Attack"

Check blocked IPs

sudo cscli decisions list

Figure 17: blocked attacker ip using crowdsec

Now test from kali using ping command

ping 10.0.2.4

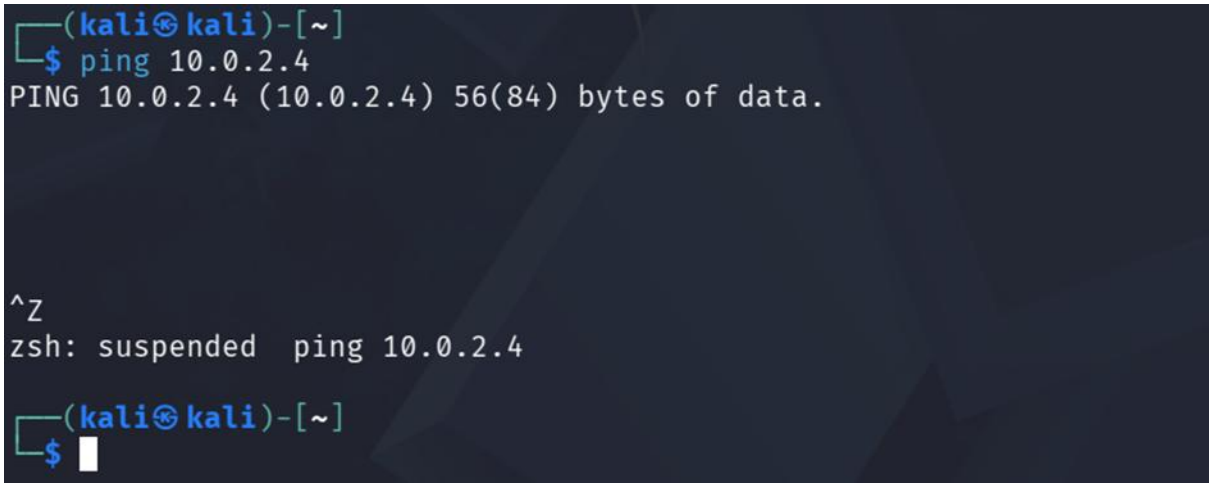

Figure 15: ping fails because crowdsec blocking the attacker IP

Blocked the malicious IP 10.0.2.15 using CrowdSec after detecting a VSFTPD attack. The block was applied through the cscli command with the action set to ban. To verify the block, a ping test was performed, confirming that the attacker's IP could no longer reach the server, ensuring immediate containment and mitigation of the threat.

Stakeholder Briefing:

A simulated attack was performed on the Metasploitable2 VM using the VSFTPD backdoor exploit. In this attack , kali linux was attacker machine which tried to exploit the know vulnerability. wazuh was detected unauthorized access attempt and triggered alert. As a response, using crowdsec blocked the attacker machine IP. Verification tests confirmed the block was successful.
This task helped to understand the attack detection, response, containment actions and importance of continuous monitoring, regular updates to protect systems.